PADHAKU
Curiosity · Class 6 Science



CHAPTER 5

Measurement of Length and Motion

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ Chapter opening

? Are the tape and rod similar to the scale that the elder sister has in her geometry box? What did mother mean by char angula?

✔ ANSWER

Yes. The rod, the tape and the geometry-box scale are all used to measure length and have similar markings (cm and mm). The tape is flexible, the rod and the scale are not.

Char angula means the width of four fingers placed side by side. Mother wanted the uniform made longer by about four finger widths.

⭐ 5.1 How do we Measure?

? Table 5.1: Different students got different numbers of handspans for the same table. Why should the number be different?

✔ ANSWER

Because the handspans of different students are of **different sizes**. A smaller handspan fits more times, a bigger one fits fewer times.

So body parts are not reliable units; we need a unit that is the same for everyone.

⭐ 5.2 Standard Units

? Would it be convenient to use the unit metre to measure larger lengths, such as the length of a railway track between two cities, or to measure smaller lengths, such as the thickness of a page of a book?

✔ ANSWER

No.

- For long lengths like a railway track between cities, the number of metres would be very large, so we use **kilometre (km)**.
- For very small lengths like the thickness of a page, a metre is far too big, so we use **millimetre (mm)** (or even smaller units).

⭐ 5.3 Correct Way of Measuring Length

? Suppose we all measure the length of the table again, but this time using a metre scale. Will the results still be different?

✔ ANSWER

The results will be the **same or nearly the same**, because a metre scale is a standard unit that is the same for everyone.

Small differences can still come if the scale is not placed correctly or the eye is not in the right position. So we must learn the correct way of using a scale.

Activity 5.1 **Measure the lengths of a comb, a pen, a pencil and an eraser using a scale and note them in Table 5.2. Compare with your friends. Are the lengths the same or slightly different? If not, discuss possible reasons.**

💡 **SAMPLE ANSWER**

Object	Length of the object
Comb	12.5 cm
Pen	14.2 cm
Pencil	17.0 cm
Eraser	4.0 cm

(Your own values will differ. Always write the number and the unit.)

Lengths of the same object measured by friends are the same or **slightly different**.

Possible reasons: scale not placed along the length of the object, eye not directly above the end (wrong eye position), starting from a broken end or not from zero, reading the mm marks wrongly.

? **Why are some length measuring devices made up of flexible materials?**

✔ ANSWER

Flexible devices (like a tailor's tape) can bend along curved surfaces. So they can measure curved lengths such as the girth of a tree, the size of the chest or the waist, which a stiff scale cannot measure directly.

⭐ 5.4 Measuring the length of a curved line

? **How would Anish and his parents have measured the required length of string lights for the arches of the verandah?**

✔ ANSWER

They could use a **flexible measuring tape** along the arches.

Or they could place a **thread** along the curved arches, mark its ends, straighten the thread and measure its length with a metre scale.

⭐ 5.5 Describing Position

? **Who do you think is correct about whether the garden or the school is closer? (Fig. 5.9)**

✔ ANSWER

All of them are correct. Each one measured the distance from his or her **own house**, so the answers are different.

If all of them had used the same reference point (say, the bus stand), their answers would have been the same.

? **What do such kilometre stones indicate? How could Padma conclude that she was getting closer to her destination?**

✔ ANSWER

Kilometre stones show the **distance of that place from a reference point**, here Delhi.

The numbers on the stones kept **decreasing** (Delhi 70 km, then Delhi 60 km). So her distance from Delhi was getting less and she knew she was getting closer.

? **Does this mean that the position of Padma, with respect to the reference point, is changing with time? When does the position of an object change with respect to a reference point? Does it change when an object is moving?**

✔ ANSWER

Yes, Padma's position with respect to Delhi is changing with time (70 km, then 60 km).

The position of an object changes with respect to a reference point when the object **moves**.

Yes, when an object is moving, its position with respect to the reference point changes with time.

⭐ 5.6 Moving Things

Activity 5.2 **List five objects in motion and five objects at rest with justification (Table 5.3). How can one decide if an object is in motion or at rest?**

💡 SAMPLE ANSWER

Objects in motion	Justification	Objects at rest	Justification
Cow grazing in the field	Its position changes with time	Tree	Its position does not change with time
Car on the road	Position changes w.r.t. the road	House	Stays at the same place
Flying bird	Position changes w.r.t. the tree	Electric pole	Stays at the same place
Running child	Position changes w.r.t. the ground	Table in class	Stays at the same place

Ceiling fan blades	Position changes w.r.t. the ceiling	Blackboard	Stays at the same place

An object is **in motion** if its position changes with respect to a reference point with time; it is **at rest** if its position does not change with time.

Think it over! **You are on a ship moving at a constant speed along a straight line on a calm sea, with no window. Is there any way you can determine whether the ship is moving or is stationary?**

💡 **SAMPLE ANSWER**

No. Inside the ship everything (you, the walls, the furniture) moves together, so with respect to things inside the ship you are at rest.

Without seeing an outside reference point (like the shore), you cannot tell whether the ship is moving at a constant speed or standing still.

⭐ 5.7 Types of Motion

Activity 5.3 **Drop an eraser from a height and observe its motion. Does it move along a straight line? When an orange drops from the tree, does it move in a straight line? Do students in a march-past move on a straight-line path?**

✔ ANSWER

Yes, the falling eraser moves straight down along a straight line.

Yes, a falling orange also moves in a straight line.

Yes, students in a march-past move on a straight-line path.

Conclusion: these are examples of **linear motion**.

? **Identify such linear motion in your surroundings.**

💡 SAMPLE ANSWER

- A car or bus on a straight road
- A train on a straight track
- A ball rolling on a flat floor
- A person walking along a straight corridor

? **You might have enjoyed playing on swings and merry-go-rounds. Are these types of**

motion also linear motion?

✔ ANSWER

No. A merry-go-round moves along a circular path (**circular motion**). A swing moves to and fro (**oscillatory motion**).

Activity 5.4 **Tie an eraser to a thread, whirl it and observe its motion. Is the motion of the eraser the same as that of a merry-go-round?**

✔ ANSWER

Yes. The eraser moves along a circular path, like a merry-go-round.

Conclusion: this is **circular motion**.

Activity 5.5 **Hang an eraser by a thread, take it slightly to one side and release. Does it start moving to and fro? Is its motion similar to the motion of a swing?**

✔ ANSWER

Yes, it moves to and fro about its fixed (rest) position.

Yes, it is similar to the motion of a swing.

Conclusion: this is **oscillatory motion**.

Activity 5.6 **Press one end of a thin metal strip to a table, press the free end slightly and**

let it go. Does it move up and down?

✔ ANSWER

Yes, the free end moves up and down again and again about its rest position.

Conclusion: this is also **oscillatory motion**.

Activity 5.7 **Observe the motions in a children's park (Fig. 5.18). Classify them as linear, circular or oscillatory motion in Table 5.4 with justification.**

💡 SAMPLE ANSWER

Object	Type of motion	Justification
Swing	Oscillatory	Moving to and fro
Merry-go-round	Circular	Moves along a circular path
Child going down a straight slide	Linear	Moves along a straight line
See-saw	Oscillatory	Moves up and down about a fixed point
Giant wheel	Circular	Moves along a circular path
Child running on a straight path	Linear	Moves along a straight line

(Check the picture in your book.)

Let us enhance our learning

1 **Match the lengths in Column I with the units suitable for measuring them in Column II.**

Column I	Column II
Distance between Delhi and Lucknow	centimetre
Thickness of a coin	kilometre
Length of an eraser	metre
Length of school ground	millimetre

✔ ANSWER

Column I	Column II
Distance between Delhi and Lucknow	**kilometre**
Thickness of a coin	**millimetre**
Length of an eraser	**centimetre**
Length of school ground	**metre**

2 (i) **The motion of a car moving on a straight road is an example of linear motion.**

✔ ANSWER

True (T). The car moves along a straight line.

2 (ii) **Any object which is changing its position with respect to a reference point with time is said to be in motion.**

✔ ANSWER

True (T). This is the meaning of motion.

2 (iii) **1 km = 100 cm**

✔ ANSWER

False (F). 1 km = 1000 m = 1,00,000 cm.

3 **Which of the following is not a standard unit of measuring length? (i) millimetre (ii) centimetre (iii) kilometre (iv) handspan**

✔ ANSWER

(iv) handspan. Its size differs from person to person, so it is not a standard unit.

4 **Search for the different scales or measuring tapes at your home and school. Find out the smallest value that can be measured using each of these scales. Record your observations in a tabular form.**

💡 SAMPLE ANSWER

Scale / tape	Where found	Smallest value it can measure
15-cm scale	Geometry box	1 mm
Metre scale	School	1 mm (some only 1 cm)
Tailor's tape	Home	1 mm (some only 0.5 cm)
Measuring tape (steel)	Home / carpenter	1 mm

Long measuring tape	Sports room	1 cm

5 **Suppose the distance between your school and home is 1.5 km. Express it in metres.**

✔ ANSWER

1 km = 1000 m

1.5 km = 1.5 × 1000 m

= 1500 m

6 **Take a tumbler or a bottle. Measure the length of the curved part of the base of the glass or bottle and record it.**

💡 SAMPLE ANSWER

Wrap a thread once around the base of the bottle. Mark the point where the thread meets its starting end.

Straighten the thread and measure up to the mark with a scale (or use a flexible tape directly).

Sample result: length of the curved base of a tumbler = **22 cm**.

7 **Measure the height of your friend and express it in (i) metres (ii) centimetres and (iii) millimetres.**

💡 SAMPLE ANSWER

Sample: height of my friend = 140 cm

(i) 140 ÷ 100 = **1.4 m**

(ii) **140 cm**

(iii) 140 × 10 = **1400 mm**

8 **You are given a coin. Estimate how many coins are required to be placed one after the other lengthwise, without leaving any gap, to cover the whole length of the chosen side of a notebook. Verify your estimate by measuring the side of the notebook and the size of the coin using a 15-cm scale.**

💡 **SAMPLE ANSWER**

Sample: Estimate = about 10 coins.

Length of notebook side = 24 cm; size (diameter) of coin = 2.4 cm (₹2 coin, check yours)

Number of coins = 24 ÷ 2.4 = **10 coins**

So the estimate is verified. (Your values may be different.)

9 **Give two examples each for linear, circular and oscillatory motion.**

✔ **ANSWER**

- **Linear:** a car on a straight road; an eraser falling from a height (or a march-past)

- **Circular:** a merry-go-round; an eraser tied to a thread and whirled (or blades of a ceiling fan)
- **Oscillatory:** a swing; a pendulum (or a metal strip pressed at one end and released)

10 **Observe different objects around you. Make a list of three objects each whose lengths are easier to express in mm, cm and m (Table 5.6).**

💡 **SAMPLE ANSWER**

Size	Objects
mm	Thickness of a coin, length of an ant, thickness of a pencil lead
cm	Pencil, eraser, notebook
m	Height of a door, length of a classroom, length of a bus

11 **A rollercoaster track is made in the shape shown in Fig. 5.19. A ball starts from point A and escapes through point F. Identify the types of motion of the ball on the rollercoaster and corresponding portions of the track.**

💡 **SAMPLE ANSWER**

- **A to B:** straight slope, so **linear motion**
- **B to C to D to E:** the round loop, so **circular motion**

- **E to F:** straight part, so **linear motion**

(Check the picture in your book.)

12 **Tasneem wants to make a metre scale. She considers plywood, paper, cloth, stretchable rubber and steel. Which of these should she not use and why?**

✔ ANSWER

She should **not use cloth, paper and stretchable rubber**.

Stretchable rubber and cloth can stretch, and paper can fold, tear or shrink, so the length of the scale would change and give wrong readings.

Plywood and steel keep their length, so they can be used.

13 **Think, design and develop a card game on conversion of units of length to play with your friends.**

💡 SAMPLE ANSWER

Sample game ("Unit Snap"):

- Make pairs of cards: one card with a length (e.g. 2 km) and another with the same length in another unit (2000 m). Other examples: 5 cm and 50 mm, 3 m and 300 cm, 1.5 km and 1500 m.
- Shuffle and place all cards face down.

- Each player turns over two cards. If they show the same length, the player keeps the pair; otherwise turn them back.
- The player with the most pairs wins.

📘 Learning further

1 **Can you find the thickness of a single page of your notebook or textbook using a scale? Think of a way and write it. Carry out the activity and report your result.**

💡 SAMPLE ANSWER

Press 100 pages (50 sheets) of the book together tightly and measure their total thickness with a scale.

Divide by the number of pages.

Sample: thickness of 100 pages = 5 mm, so thickness of one page = 5 ÷ 100 = **0.05 mm**.

2 **Collect fallen leaves from the same tree, measure their length and breadth with a 15-cm scale and record in Table 5.7. Discuss why the leaves of the same tree vary in length and breadth.**

💡 SAMPLE ANSWER

S. no.	Name of tree	Length of leaf	Breadth of leaf
1.	Peepal	12.0 cm	8.5 cm

2.	Peepal	10.5 cm	7.8 cm
3.	Peepal	13.2 cm	9.0 cm

The leaves vary because they are of different ages (young and old), and they get different amounts of sunlight, water and food on different parts of the tree.

3 **Discuss with elders what units were used for measurement of length in the olden days. Also, using the internet, find out about the length scales found in excavations of archaeological sites in India.**

💡 **SAMPLE ANSWER**

Old units: angula (finger width), balisht (handspan), haath (arm length from elbow to fingertip), kadam (stride), gaz, foot, dhanusa and yojana.

Objects with ruled markings that could be scales have been found at sites of the **Harappan Civilisation** (for example, a scale piece found at Lothal in Gujarat).

4 **Create a maze using lines of 1 cm, 2 cm and their combination. Use your imagination and expand it (Fig. 5.21).**

💡 **SAMPLE ANSWER**

Draw a square boundary on paper. Using a scale, draw straight walls of 1 cm, 2 cm, 3 cm (1 + 2) etc. inside it, leaving gaps for paths, with one entry and one exit. Check that there is only one correct path. Try it with friends.

5 **How tall am I? Stand along a wall and, with the help of an adult, mark your height. Repeat it every three months to maintain a height record.**

💡 **SAMPLE ANSWER**

Date	My height	Sibling's height
January	135 cm	120 cm
April	136 cm	121.5 cm
July	137.5 cm	123 cm

The record shows that we grow taller over time. (Use your own values.)

6 **Measure the distance between two places using a bicycle (count the turns of the wheel and multiply by the length of the outer boundary of the wheel). Find out about a 'Jones Counter'.**

💡 **SAMPLE ANSWER**

Sample: outer boundary of the wheel = 2 m; number of sounds (turns) = 250

Distance = 2 m × 250 = **500 m**

> A **Jones Counter** is a small counting device fixed to the front wheel of a bicycle. It counts the turns of the wheel, so it is used to measure the exact distance of road-running race courses.

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**